

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology

Department of Electrical & Electronic Engineering



Assignment

Course Code: EEE 4167

Course Title: Optoelectronics

Submitted by:

Name: Mir Masrur

Roll: 2001019

Section: 'A'

Year: 4th Year (ODD)

Submitted to:

Md. Khayrul Islam

Lecturer

Department of Electrical &

Electronic Engineering, RUET

Date of Submission: 29-10-25

Problem Statement

There are several techniques to work in the field of light modulation. Like the electro-optic effect, magneto-optic effect, or acousto-optic effect. For my situation I will describe an electro-optic modulation, which works based on the electro-optic effect. This phenomenon occurs when the refractive index of certain crystalline materials changes in response to an applied electric field. Among the two main types of electro-optic effects, the Pockels effect which is a linear effect and the Kerr effect that is quadratic in nature, this work focuses on the Pockels effect due to its linear response, high speed, and suitability for practical devices.

The electro-optical effect refers to the change in refractive index caused by the modulation of free-carrier concentration in a material, which results in the absorption of light and a corresponding change in the phase of light passing through a waveguide.[1]

When an electric field is applied across an electro-optic crystal like lithium niobate (LiNbO_3) or potassium dihydrogen phosphate (KDP), the refractive index changes in proportion to the field strength. This change introduces birefringence, which alters the phase and polarization of transmitted light. By controlling the applied voltage, the phase of the optical wave can be varied continuously. With the right setup of polarizers and analyzers, this phase modulation can be turned into intensity modulation that a photodetector can detect.

Electro-optic modulation is very useful because for its-

- High-speed signal transmission in optical fiber communication systems with tens of GHz bandwidth.
- Precise phase control in interferometric sensors and laser systems.
- Efficient optical switching and amplitude modulation in integrated photonic circuits.

Thus, the electro-optic modulator acts as an important link between the electrical and optical systems, allowing for the controlled manipulation and detection of light signals in optoelectronic applications.

Proposed Model

Laser/LED $\rightarrow$ Polarizer $\rightarrow$ Electro-Optic Crystal $\rightarrow$ Analyzer $\rightarrow$ Photodetector

Here,

- The polarizer makes light linearly polarized.
- The electro-optic crystal (e.g., KDP, LiNbO_3) experiences birefringence under voltage.
- The analyzer converts phase/polarization changes into measurable intensity changes.
- The detector senses modulated light intensity.

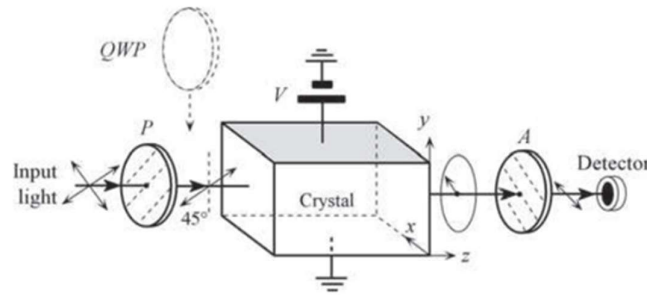


Fig.1: An overview of a setup a detector of light modulation using electro-optic effect [2]

As shown in the fig.1 a light source is going through the polarizer that converts the light source into a plane wave. Then it enters the electro-optic crystal. There are different types of crystal can be used such as KDP, LiNbO₃ etc [3]. And along the crystal a external voltage source is applied to change the formation of electron into the crystal. That change the refractive index of the material. Which change the phase of light that can be detected in the detector. This is the simple explanation on how this set up works.

So,

Applied Electric field → Change refractive index, Δn → Change the phase, $\Delta\phi$ → That create light modulation

1. General expression for change in refractive index:

$$\Delta n = rE_0E + pE_0E^2$$

Where,

E =Applied electric field

r =Linear electro-optic coefficient → Pockels effect

p =Quadratic electro-optic coefficient → Kerr effect

The first term of the equation is the result of Pockels effect and the second term is for Kerr effect. [4]

Table 1: Material Examples & EO Coefficients

Material	EO Coefficient (r)	Type
KDP (KH ₂ PO ₄)	10.6 pm/V	Linear EO
ADP (NH ₄ H ₂ PO ₄)	8.5 pm/V	Linear EO
Lithium Niobate (LiNbO ₃)	30.8 pm/V	Strong EO
Lithium Tantalate (LiTaO ₃)	30.3 pm/V	Strong EO

2. Linear Electro-Optic Relation — *Pockels Effect*:

For materials showing only linear response:

$$n(E) = n_0 \pm \frac{1}{2} n_0^3 r E = n_0 \pm \frac{1}{2} n_0^3 r \frac{V}{L}$$

Where,

n_0 = Refractive index with no field

V = Applied voltage

L = Length of crystal

The $\pm$ indicates that refractive index can increase or decrease depending on crystal orientation and field polarity.

3. Resulting Phase Shift:

$$\Delta\phi = \frac{2\pi}{\lambda} n_0^3 r \frac{VL}{2d}$$

This is equivalent to:

$$\Delta\phi = \frac{\pi V}{V_\pi}$$

where the half-wave voltage is:

$$V_\pi = \frac{\lambda d}{n_0^3 r L}$$

V_π = voltage needed to produce a phase shift of π radians (half a wavelength).

Graphical Representation

To visualize the theory, we generated three graphs using MATLAB:

1. Phase Shift vs Voltage: Shows linear increase ($\Delta\phi \propto V$)
2. Output Intensity vs Voltage: Follows $\sin^2$ relationship
3. Modulation Depth vs Frequency: Shows low-pass behavior with cutoff frequency $f_c = 1/(2\pi RC)$

These plots confirm that as voltage increases, phase shift rises linearly, intensity oscillates periodically, and modulation depth reduces at higher frequencies. Together, they help us understand how voltage directly affects optical output.

Assumed parameters,

$\lambda = 1.55 \times 10^{-6}$; Wavelength of light, let this be $1.55 \mu\text{m}$

$n = 2.2$; Refractive index of LiNbO₃ crystal

$r = 30.8 \times 10^{-12}$; Electro-optic coefficient (Pockels coefficient), let this be 30.8 pm/V

$L = 1 \times 10^{-2}$; Length of the crystal, let this be 1 cm

$d = 5 \times 10^{-4}$; Electrode spacing, let this be 0.5 mm

$I_{\text{in}} = 1$; Input optical intensity, assume normalized to unity.

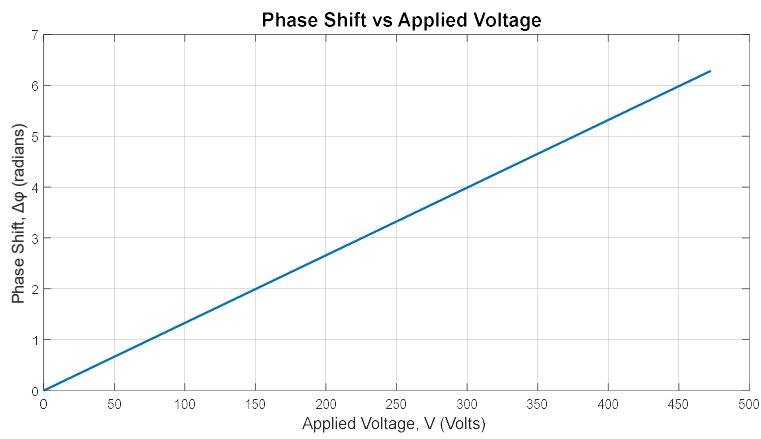


Fig.2 : Changing phase in response to applying voltage

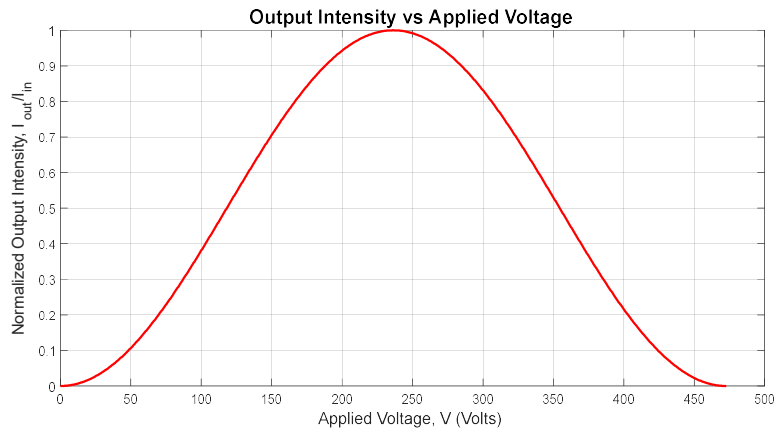


Fig.3: Relation between output intensity and applied voltage

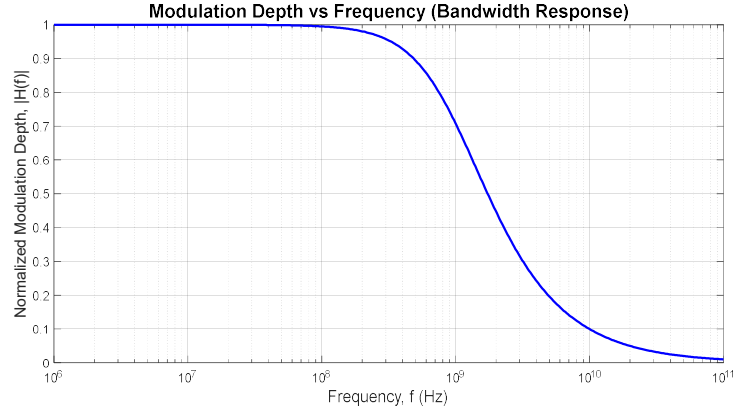


Fig.4: Modulation depth in response to frequency

Results and Discussion

In this section, we present how the applied electric field affects the light passing through the electro-optic modulator-

1. Relationship Between Input Electric Field and Optical Response

The electro-optic (EO) effect is governed by the change in refractive index with respect to the applied electric field:

$$\Delta n = -\frac{1}{2}n^3rE$$

where

n = refractive index without field,

r = linear electro-optic (Pockels) coefficient,

$E = V/d$ = applied electric field (V : voltage, d : electrode spacing).

This refractive index variation introduces a phase shift in the propagating light:

$$\Delta\phi(V) = \frac{2\pi}{\lambda}\Delta nL = -\frac{\pi n^3rL}{\lambda d}V$$

Where,

λ = wavelength of light,

L = crystal length.

2. Theoretical Result Description

The half-wave voltage, which produces a phase shift of π radians, is given by:

$$V_\pi = \frac{\lambda d}{n^3rL}$$

This is a key performance parameter that determines modulation efficiency.

When the EO crystal is placed between a polarizer and analyzer, the modulated light intensity after transmission can be expressed as:

$$I_{out}(V) = I_{in} \sin^2\left(\frac{\Delta\phi(V)}{2}\right)$$

Substituting $\Delta\phi(V) = \pi V/V_\pi$, we get:

$$I_{out}(V) = I_{in} \sin^2\left(\frac{\pi V}{2V_\pi}\right)$$

This relation shows a $\sin^2(V)$ dependence, meaning that light intensity oscillates between zero and its maximum value as voltage increases.

Discussion on Performance Parameters

We know that the performance of an EO modulator depends on three key parameters: modulation depth, bandwidth, and response time.

a) **Modulation Depth [5] (m):**

Modulation depth defines the degree of light intensity variation due to modulation:

$$m = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$$

For complete phase swing ($\Delta\phi = \pi$), $m = 1$, corresponding to 100% modulation.

High modulation depth improves the signal-to-noise ratio (SNR) and enhances detection accuracy.

b) **Bandwidth [6] (f_n):**

The EO effect itself is nearly instantaneous, but the device's electrical circuit limits its operating bandwidth:

$$f_c = \frac{1}{2\pi RC}$$

where R and C represent the driver resistance and electrode capacitance.

For integrated EO modulators, typical bandwidths exceed 10–40 GHz, ensuring faithful signal detection in optical communication systems.

c) **Response Time (τ):**

The response time, the inverse of bandwidth, dictates how quickly the modulator reacts to input voltage changes:

$$\tau = \frac{1}{2\pi f_c}$$

A shorter τ means faster modulation and higher detection accuracy.

For $f_c = 1$ GHz, the response time is approximately 160 ps, sufficient for high-speed optical links.

3. Practical Implications and Real-World Applications

In practical optical systems, we can use this electro-optic modulation principle for several purposes, such as -

(a) Optical Communication:

High-speed data transmission relies on encoding electrical signals into optical carriers using EO modulators.

Intensity modulation follows the relation:

$$I_{out}(t) = I_{in} \sin^2\left(\frac{\pi V(t)}{2V_\pi}\right)$$

allowing voltage-driven modulation of light for digital or analog communication.

(b) Optical Sensors:

In interferometric sensors, changes in environmental parameters (strain, pressure, temperature) alter the applied voltage or field, producing a measurable phase shift:

$$\Delta\phi = \frac{\pi n^3 r L}{\lambda d} V_{sensor}$$

The detected intensity variation provides a precise measure of the physical stimulus.

(c) Laser Systems:

EO modulators are used for Q-switching and pulse modulation of lasers. Rapid phase or amplitude control through:

$$E(t) = E_0 \sin(\omega t + \Delta\phi(V))$$

enables shaping of laser pulses and stabilization of optical resonators.

(d) Integrated Photonics:

In LiNbO₃ or silicon-photonics chips, the same EO relations are used to design on-chip modulators with low drive voltages and high speed. The linear dependence of phase on voltage simplifies electronic control and integration with CMOS circuits.

Addressing CLO2 and PLO3

We have demonstrated how this system detects and controls light through electrical modulation. The applied voltage changes the refractive index of the EO crystal as:

$$\Delta n = -\frac{1}{2} n^3 r \frac{V}{d}$$

which changes the output intensity:

$$I_{out}(V) = I_{in} \sin^2\left(\frac{\pi V}{2V_{\pi}}\right)$$

We can see that by detecting these changes with a photodetector, we effectively measure light intensity variations caused by electrical input. Hence, we achieve light detection through modulation, fulfilling CLO2. The process also demonstrates analytical design and evaluation of a system, which satisfies PLO3, as we design, analyze, and optimize a real-world optoelectronic mechanism.

Addressing Complex Engineering Problems (P1–P3)

Attribute	Justification in the Proposed Electro-Optic System
P1: Depth of Knowledge Required	We need to understand the Pockels effect, birefringence, and how electric fields affect light propagation.
P2: Range of Conflicting Requirements	We must balance modulation efficiency, voltage requirements, and bandwidth to optimize performance.
P3: Depth of Analysis Required	We analyze the relation between voltage, refractive index, and light intensity using derived equations.

Conclusion

In this work, we have discussed the concept and operation of electro-optic modulation. We observed that applying an external voltage changes the crystal’s refractive index, which then alters the phase and intensity of light. By using materials like LiNbO₃ and KDP, we can achieve high-speed modulation with precise control. We also studied how modulation depth, bandwidth, and response time affect system performance. Overall, we can conclude that the electro-optic modulator provides an efficient way to control and detect light in modern optical communication and photonic systems.

References

- [1] *ScienceDirect Topics*, “Electro-Optical Effect,” Elsevier B.V., 2024. [Online]. Available: <https://www.sciencedirect.com/topics/physics-and-astronomy/electro-optical-effect>.
- [2] S. O. Kasap, *Optoelectronics and Photonics: Principles and Practices*, 2nd ed. Pearson, 2013.
- [3] *PM Optics*, “Electro-Optical Crystals,” PM Optics, 2024. [Online]. Available: https://www.pmoptics.com/electro-optical_crystals.html
- [4] B. E. A. Saleh and M. C. Teich, *Fundamentals of Photonics*, 3rd ed., Hoboken, NJ: Wiley, 2019, ch. 18, pp. 945–970.
- [5] T. Kawanishi, *Electro-Optic Modulation for Photonic Networks: Precise and High-Speed Control of Lightwaves*, Cham, Switzerland: Springer, 2022.
- [6] S. Hou, W. Chen, X. Yang, and Z. Zhou, “High-Speed Electro-Optic Modulators Based on Thin-Film Lithium Niobate,” *Nanomaterials*, vol. 14, no. 10, pp. 1-18, 2024.